

Candidate Instruments for a Universal Carbon Premium: The Commonwealth Cost of Carbon

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1 Abstract

This paper examines how to develop a reasonable contractual proposition for a society to cooperate fairly in the interest of a common ecosystem. It recognises that no society is entirely free from political realities or the costs involved in maintaining social order. As a result, ideal propositions of social cooperation, such as The Law of Peoples, remain incomplete. A method of price discovery is required for reasonable individuals to cooperate in the interest of the commons. Previous studies have attempted to discover these prices through application of Frank Ramsey's Mathematical Theory of Saving. However, we introduce the Commonwealth Cost of Carbon (CCC), as a new Ecosystem Performance Observation that measures how much societies spend to maintain order relative to their total activity. This ratio provides a transparent, elegant and replicable way to assess global ecosystem performance without resorting to those with privileged access. Using public accounts, CCC were estimated from 2000 to 2020. Over this relatively short period, the CCC increased from $\sim 2000\$74$ to $\sim 2020\$154$ per tonne of CO₂e, suggesting both a decline in ecosystem performance and a rising common value for emission savings. The range of uncertainty in these point estimates was as much as ± 25 percent in the year 2000, with a trend for improved observation accuracy since. As an observation, the CCC provides a transparent and replicable measure of ecosystem performance to bring institutions to account. As a natural method of price discovery, the CCC offers a basis for voluntary worldwide cooperation in ecosystem management.

Keywords: Ecosystem Performance; Environmental Economics; Social Contract; Price Discovery

2 Introduction

We live in an age where global crises such as climate change, biodiversity loss, and geopolitical instability overlap and compete for attention. Yet, our tools for cooperation are often inadequate. Such unrest raises a deeper question: how to develop a reasonable contractual proposition for a society to cooperate fairly in the interest of a common ecosystem?

For centuries, those who study ethics have debated how communities balance self-interest with duty to the wider world. Their ideas provide language for rethinking today's governance

of our common ecosystem. John Rawls, for example, argued in *The Law of Peoples* that "*Peoples have a duty to assist other Peoples living under unfavourable conditions that prevent their having a just or decent political and social regime.*" [Rawls, 1999]. Yet, in truth, everyone today lives under some form of unfavourable condition that reflects the natural limitations in social development. This implies a shared duty to sustain the systems that make our lives worth living, whatever perspective one takes on such a diverse and complex issue. So, is it realistic to imagine an entirely reasonable society unaffected by the imperfect Realpolitik of nation-states?

Philosophers describe the "*State of Nature*" as life without shared rules: sometimes cooperative [Locke, 1967] [Nozick, 1974], sometimes chaotic and conflict-ridden [Hobbes, 1668]. If our global order today resembles such a State, what responsibilities do we still owe one another? Should these duties fall primarily to nation-states, to individuals, or to organisations? What method of price discovery might provide a basis for reasonable individuals to cooperate in the interest of the global commons?

3 Perspectives and Outcomes

It is difficult to imagine any society free from the political realities and territorial disputes that shape nation-states. Even the most reasonable communities must often comply with political decisions they neither designed nor fully endorsed. These compromises create what economists call ecosystem externalities or losses: resulting in social costs that fall outside the accounting of consumers. For societies to act in the common interest, such externalities need to be recognised and internalised within decision-making [Coase, 1960]. Therefore, an Ecosystem Performance Observation becomes a necessary instrument of any "*best-we-can*" social contract: a way that develops trust, credibility, and continual learning from the state of our common ecosystem.

At its core, observing ecosystem performance means examining the consequences of human activity. Every transaction, service, or agreement produces material and energetic by-products, such as: waste, water quality, and greenhouse-gas emissions. While many forms of waste can be managed locally by territorial authorities, greenhouse gases are different: once released, they enter common planetary sinks. Emissions from many locations can most affect quite distantly related individuals via shared resources. Therefore, one might suggest that national accounting alone does not provide a fair or reasonable basis for cooperation. A universal approach is required.

Further, as carbon-based lifeforms, anthropogenic greenhouse gas emissions have deep rooted dependencies with human activity. This makes greenhouse gas emissions a particularly strong candidate for a common signal that relates human influence on the overall state of the global commons. Earth's carbon sinks, its atmosphere, forests, oceans, and soils, absorb greenhouse gas emissions and support the carbon cycle. The condition of these sinks is itself a performance indicator: their capacity is pivotal in determining mean surface temperature, which influences conditions of the wider ecosystem.

Measuring economic output alone cannot capture this reality. A society that grows while overloading these sinks cannot be said to perform well. In order to respond reasonably, one needs to recognise that these productive assets have a crucial relationship with *Enabling Assets*: institutions, knowledge, networks and time. These human-made Enabling Assets shape social production and observable behaviour and can be envisaged as enabling software that guides ecosystem development [Dasgupta, 2015]. Poorly configured Enabling Assets could have

damaging consequences for our common ecosystem, ultimately manifested in the observed performance of our carbon sinks.

Economic growth may remain a legitimate goal, particularly between struggling powers. Yet, if pursued without regard to the overall condition of our shared ecosystem, could become self-defeating. Growth that undermines the very foundations of our ecological life support systems is destabilising. A more telling measure of progress, therefore, might be the health of our carbon sinks and the effectiveness of our Enabling Assets in maintaining them.

3.1 Utilitarian Position

Utilitarians tend to prescribe happiness as a whole as the ultimate end to action [Sidgwick, 1877]. Many economists have followed this perspective to quantify environmental damage by estimating a "*Social Cost of Carbon*".

Normative economists often regard only individual circumstance, or the level of welfare in a given state of affairs, as important. This is known as the neutrality assumption [Dasgupta, 2001]. Arrow, May, and Sen argue that this neutrality implies a demand for anonymity among social states; each person's welfare must carry equal weight in the social calculus [Arrow, 1963] [May, 1952] [Sen, 1970]. Waldron extends this reasoning, defining social welfare as a goal-based aggregation of individual well-being [Waldron, 1984], while Dworkin notes that such goals become non-individuated political aims rather than expressions of rights [Dworkin, 1978]. Together, these theorists give utilitarianism its enduring attraction: it seems to offer a mathematically elegant path to fairness through aggregation.

A decision-making tool to quantify such ends was first given shape by Frank Ramsey in A Mathematical Theory of Saving [Ramsey, 1928]. This was later employed in complex integrated assessment models of humanities social and environmental systems that apply gross assumptions of Earth's development. Mechanisms that determine the state of the world are identified and the consequences of alternative policies charted so that consequences can be valued. Welfare surpluses are then estimated for policy options, by making projections of differences from the *status quo* hundreds of years into the future. Valuation of these surpluses involves setting a global social discount rate and this requires a set of personal assumptions. As a result, these economists offer estimates of a Social Cost of Carbon [Dasgupta, 2001]. In these studies, welfare over time is expressed as an infinite series of discounted utilities:

$$V_t = \sum_t^{\infty} \beta^{(\tau-t)} \cdot U(C_{\tau}) \quad \text{for} \quad t \geq 0 \quad (1)$$

Where,

$$\beta = \frac{1}{(1 + \mu)} \quad (2)$$

Where, V_t is a *generations welfare*, β is the *discount factor*, U is *welfare*, C_{τ} is *consumption during time-step, t is time* and μ the *social discount-rate*.

$$\mu = \sigma \cdot g + \delta \quad (3)$$

Where, σ is the *marginal utility of consumption* and the difference in the utility one would gain from a unit of consumption by those of low and high incomes, g is the *long-term growth rate* in consumption, δ is the *pure rate of time preference* and our impatience to consume in fear of extinction.

$$S = \frac{\mu - \delta}{\sigma \cdot \mu} \quad (4)$$

Where, S is the *savings rate* and the proportion of output that should be invested.

This deceptively simple equation conceals deep ethical assumptions. Every variable requires a normative judgment: a higher (δ) implies impatience at the expense of future generations, while a higher (σ) places greater weight on redistributing welfare from rich to poor. These are ethical choices disguised as mathematical constants.

To illustrate the sensitivity of these assumptions, [Cline, 1992], [Nordhaus, 1994], and [Stern, 2006] each applied similar underlying decision-making theory. However, they each arrived at dramatically different conclusions, as evident in Table 1. All adopted a similar headline discount rate (~ 0.05), but diverged in their treatment of equality (σ), growth (g), and time preference (δ).

Table 1: Notable Precedents: Assumptions in Social Discounting

Contribution	μ	σ	g	δ	S
[Cline, 1992]	0.05	1.5	0.033	0	0.67
[Nordhaus, 1994]	0.05	1	0.02	0.03	0.4
[Stern, 2006]	0.05	1	0.049	0.001	0.98

A comparison by the U.S. Government [Interagency Working Group on Social Cost of Carbon, 2013], found that the DICE, PAGE, and FUND studies returned a range between $<2007\$0\text{tC}^{-1}$ to a 95th percentile of $2007\$128\text{tC}^{-1}$ [Nordhaus, 1994] [Hope et al., 1993] [Tol, 1997]. The range largely reflects differences in ethical stance rather than data, exposing how personal such views can be.

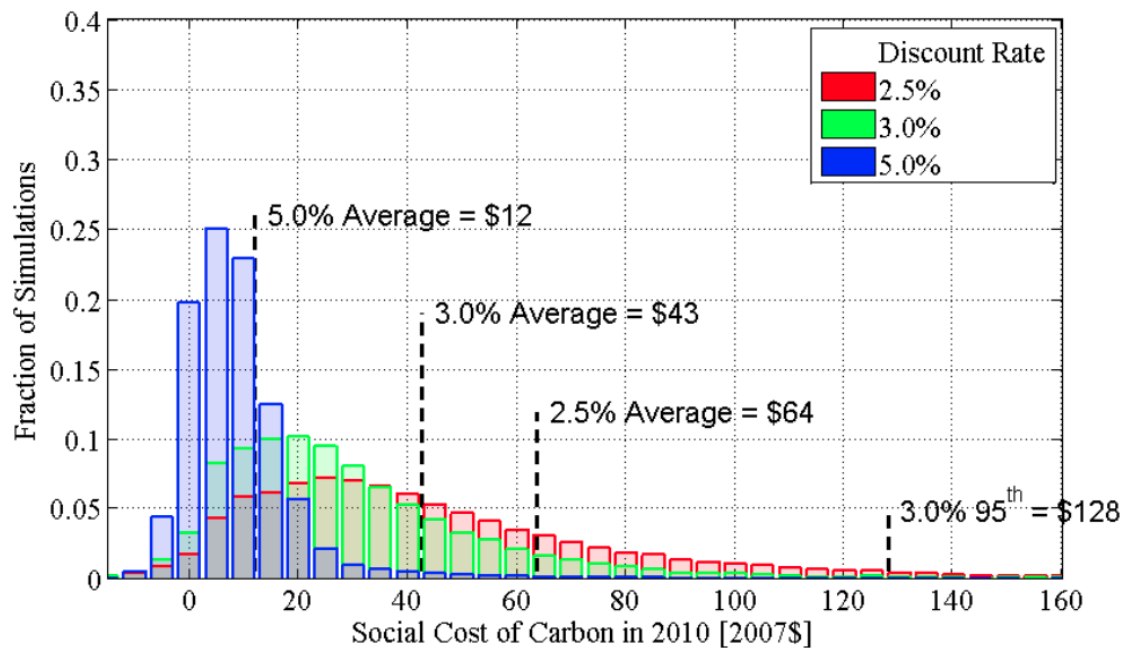


Figure 1: Social Cost of Carbon in 2010 (in 2007 dollars per metric ton)

Figure 1 shows the range of Social Cost of Carbon estimates derived from integrated assessment models. Despite their sophistication, the results differ by more than an order of magnitude, reflecting underlying value judgments rather than objective consensus. A reliance on an ideal observer renders it vulnerable to subjective bias. Each model embeds ethical positions that compete rather than converge, assuming a single planner can prescribe values for all.

Critics such as Dasgupta and Sen argue that portraying imperfect economies as though they were initial utopias in this way is a categorical error [Sen, 1997]. Even among welfare economists, some reject applying pure time preference so that one allows the living to take advantage of their position in time to favour their own interests [Sidgwick, 1877] [Ramsey, 1928]. Such examples ignore the possibility for the living to disadvantage their predecessors or descendants. Libertarian's might argue that such collective action would lead to violations of individual rights to even a slight extent and therefore should be rejected [Nozick, 1974]. Free-market advocates may criticise such methodologies for relying on centralised long-term planning, rather than allowing local agents to adjust their activities according to present situations [Hayek, 1944].

Those of the liberal tradition would consider nothing sacrosanct about the public decision concerning the level of savings and its bias with respect to time preference deserves no special respect [Rawls, 1971]. Consequently, although the Social Cost of Carbon remains a useful accounting device, its dependence on contested assumptions limits its ability to provide a common basis for price discovery. To find such a tool, we must look beyond aggregate welfare and consider the terms of cooperation under imperfect authority, toward a Hobbesian understanding of the global commons.

3.2 Hobbesian Position

Where utilitarian models seek stability through aggregated welfare, a Hobbesian perspective begins from a different premise: that stability itself is precarious. In *Leviathan*, Hobbes describes order, not as the product of shared virtue, but as an agreement born from mutual vulnerability [Hobbes, 1668]. Individuals surrender certain freedoms in exchange for collective security. This vision, of imperfect actors cooperating under constraint, makes Hobbes’s logic strikingly relevant to the global commons. Where no single sovereign authority exists to govern shared resources, yet cooperation remains essential for survival.

If the utilitarian question is *“what makes us happiest?”*, the Hobbesian question is *“what keeps us from collapse?”* Under such a framework, the cost of maintaining order becomes a measurable proxy for the cost of sustaining life itself. To capture this relationship, we define the Commonwealth Cost of Carbon (CCC) as a ratio between the resources societies devote to enforcing order and the greenhouse gases they emit. Greenhouse gas emissions are expected to be a reasonable proxy for overall activity.

“The mutuall transferring of Right, is that which men call CONTRACT”

“He that transferreth any Right, transferreth the Means of enjoying it, as farre as lyeth in his power. As he that selleth land, is understood to transfer the Herbage, and whatsoever grows upon it; Nor can he that sells a Mill turn away the Stream that drives it. And they that give to a man the Right of government in Sovereignty, are understood to give them the right of levying mony to maintain Souldiers; and of appointing Magistrates for the administration of Justice.” [Hobbes, 1668].

$$\frac{\text{anthropogenic expenditure on enforcement}}{\text{anthropogenic greenhouse gas emissions}} \quad (5)$$

This expression links moral duty to material accounting. It estimates how much humanity collectively sacrifices to enforce the contracts that uphold a life worth living. It reflects both the economic and ethical costs of sustaining the commons under conditions of imperfect cooperation.

As a worked example, we have determined point estimates for the CCC in Reporting Years 2000 to 2020. Worldwide greenhouse gas emissions were collected from the World Banks published World Development Indicators in 2024 [World Bank, 2024]. Expenditures by Reporting Nations on defense, public order and safety were retrieved from the International Monetary Fund Data Portal in 2024. National and worldwide accounts of gross domestic product were also retrieved from this data source [International Monetary Fund, 2024]. Worldwide enforcement expenditures were inferred for each Reporting Year by dividing worldwide gross domestic product by the sum of Reporting Nations gross domestic product, before multiplying the resulting ratio by the sum of Reporting Nations expenditures on defense, public order and safety. This inference with incomplete data introduces the only source of uncertainty within this calculation, which has been found to be as much as 25 percent for any given reporting year.

A CCC has been calculated for each Reporting Year, with the results presented in Table 2. Point estimates for the CCC in current prices are $\sim 2020\$154\text{tCO}_2\text{e}$, up from $\sim 2000\$74\text{tCO}_2\text{e}$.

Whilst, the uncertainty identified in the Utilitarian position largely arises through the assumptions adopted by the observer. Uncertainty in this CCC arises principally from incomplete national accounts, which we find to be both more accurate and more auditable. The table demonstrates a trend of degrading ecosystem performance and greater certainty over the Reporting Years. Such insight appears to suggest that action to save greenhouse gas emissions have become more valuable since the turn of the millenium.

Table 2: Commonwealth Costs of Carbon Summary

Year	Worldwide Enforcement \$bn	Uncertainty $\pm\%$	Worldwide Emissions MtCO ₂ e	CCC \$tCO ₂ e
2000	2796.34	0.25	37583.8	74.4
2001	2955.72	0.25	36622.8	80.71
2002	3251.24	0.24	38022.07	85.51
2003	3705.49	0.23	39701.91	93.33
2004	4184.29	0.23	42190.19	99.18
2005	4392.20	0.14	42117.70	104.28
2006	4713.21	0.14	44401.01	106.15
2007	5073.57	0.17	44544.53	113.90
2008	5861.22	0.13	44000.69	133.21
2009	5929.06	0.11	43190.55	137.28
2010	6228.25	0.10	45239.77	137.67
2011	6844.87	0.14	48725.83	140.48
2012	6814.93	0.12	49442.73	137.83
2013	6834.78	0.12	49165.56	139.02
2014	6875.71	0.12	49894.71	137.80
2015	6413.59	0.12	50486.71	127.04
2016	6452.82	0.13	49124.78	131.36
2017	6780.93	0.12	50599.36	134.01
2018	7338.97	0.16	5288.78	140.35
2019	7528.97	0.16	52602.35	143.13
2020	7713.93	0.15	50068.76	154.07

The indicator's simplicity is its strength: anyone can compute it using open data, without recourse to privilege. A transparent accounting of how much societies pay, in the broadest sense, to sustain living conditions within natural disorder. Without the penal machinery of the state, Hobbes warned, humanity would risk *"a war of every man against every man"*. In this context, the CCC quantifies the ongoing sacrifice that prevents such collapse, the expenditure that buys stability. Carbon emissions thus appear not only as pollutants but as signatures of human interdependence; their true cost includes the price of sustaining peace. It does not presume altruism, nor rely on an omniscient planner. Instead, it measures cooperation under constraint. The CCC therefore offers a different basis for a universal carbon premium. Rather than prescribing what carbon ought to cost through assumptions about future welfare, it discovers a common reference value from observable expenditure and emissions. The resulting premium could provide a consistent basis for cooperation between individuals and organisations operating across transnational boundaries.

4 Conclusions

This research began with a question that defines our age: how to develop a reasonable contractual proposition for a society to cooperate fairly in the interest of a common ecosystem? No community is exempt from the political realities or costs of maintaining order. Ideal visions of justice, such as Rawls's Law of Peoples, remain incomplete when confronted with ecosystem interdependence and asymmetries of power. The challenge, therefore, is to connect philosophical reasoning with measurable practice.

The CCC emerged from this effort as a transparent, auditable Ecosystem Performance Observation that also provides a method of price discovery for transnational cooperation. It measures how much humanity collectively invests in maintaining peace, order, and stability per tonne of greenhouse gas emitted. Between 2000 and 2020, this cost rose from $\sim 2000\$74$ to $\sim 2020\$154$ per tonne of CO₂e, revealing both a decline in ecosystem performance and a rising social value for emission savings. The range of uncertainty in these point estimates was as much as ± 25 percent in the year 2000, with improved observation accuracy since. Unlike previous Social Cost of Carbon models, the CCC does not depend on hidden ethical parameters or privileged access. It can be calculated by any actor, anywhere, using open data, translating moral duty into a common metric.

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